

Valve Spring Finite Element Analysis

Drawing A177 053 05 00 | Intake Valve Spring (Beehive)

Spring Geometry

Free length:	46.1 mm
Wire cross-section:	2.92 × 3.66 mm (oval)
Total coils:	8.6 (1.25 closed each end)
Active coils:	6.1
OD range:	19.6 → 15.7 mm (beehive)
Pitch range:	4.96 → 7.76 mm (top→bot)
Solid height:	≈ 23.6 mm

Operating Conditions (Drawing / Measurement)

Installed length (draw.):	36.1 mm
Installed leng	

CAD Model (STEP)
Valve Spring A1770530500
Beehive, Oval Wire 2.92x3.66mm, L0=46.1mm

Preload force

Valve lift:

Full-lift force (

Material:

FE Model S

Solver:

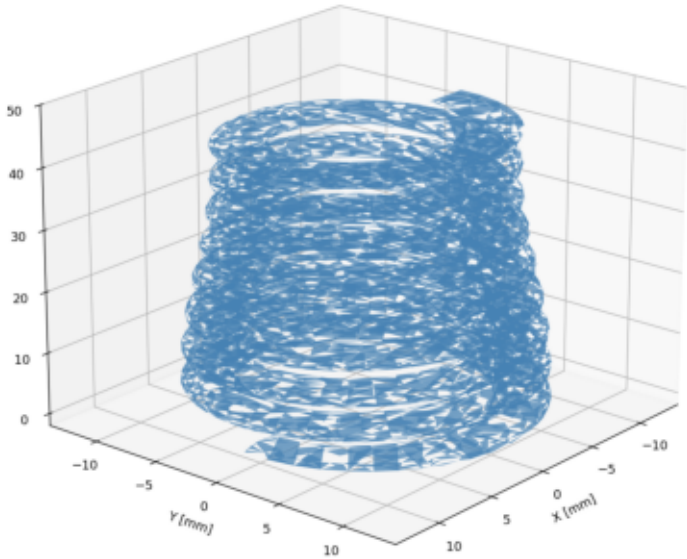
Elements:

Nodes:

Analysis:

Contact: Self-contact, HARD+PENALTY

Steps: 2 (preload → valve lift)



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Analysis Steps

Step 1 — Assembly Preload

Compress 46.1 → 36.1 mm
(s = 10.0 mm from free length)
Achieved: 244 N (drawing: 250 N)
NBOT: fully fixed (UX=UY=UZ=0)
NTOP: UX=UY=0, UZ=-10.0 mm
NLGEOM, self-contact active

Step 2 — Valve Lift

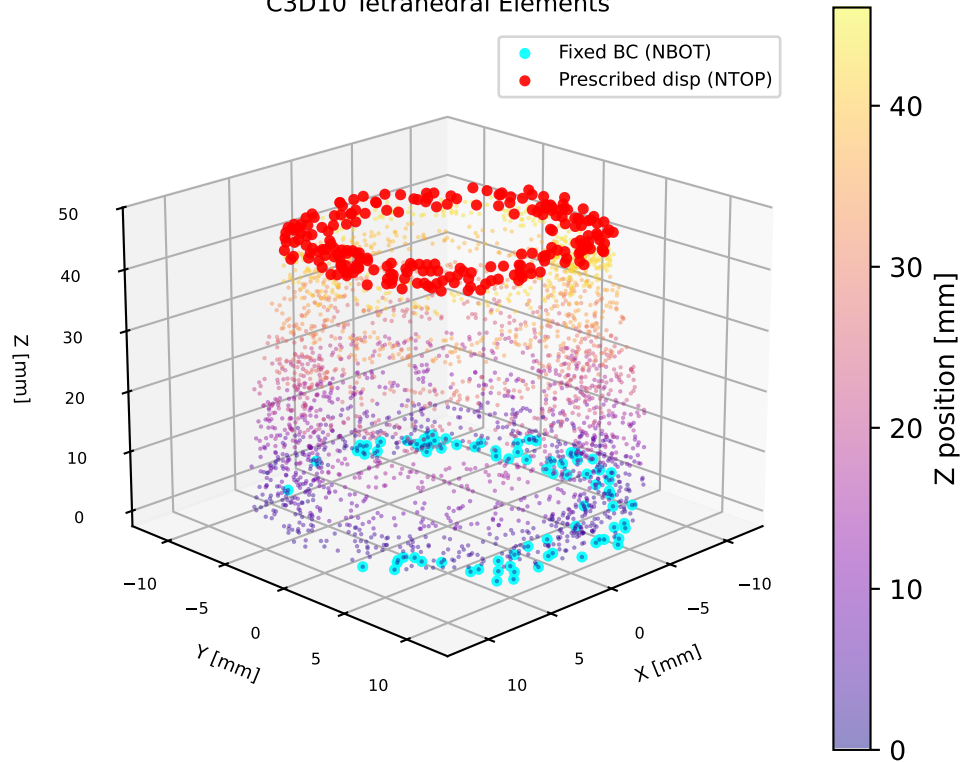
Continue: 36.1 → 26.1 mm
(additional 10 mm valve lift)
Achieved: 798 N (measurement: 621 N)
NBOT: fully fixed (OP=NEW)
NTOP: UZ=-20.0 mm (total)
NLGEOM, progressive stiffening

Self-Contact Setup

Surface type : ELEMENT (exterior free faces)
Contact zone : z = 4.15 – 41.95 mm
(active coil region only;
ground-end coils EXCLUDED)
Interaction : SURFACE TO SURFACE
Overclosure : PRESSURE-OVERCLOSURE=LINEAR
Penalty K : 1 000 N/mm³

Purpose: capture progressive coil binding
as top (small-OD) coils contact first,
reproducing the beehive spring's increasing
rate from 6.1 → ~3.1 active coils over
the 10 mm valve lift stroke.

FE Mesh — 54,529 Nodes
C3D10 Tetrahedral Elements



Valve Spring Force vs Lift — FEA vs Measurement
Free length 46.1 mm | FEA installed 36.1 mm | Valve lift 10 mm

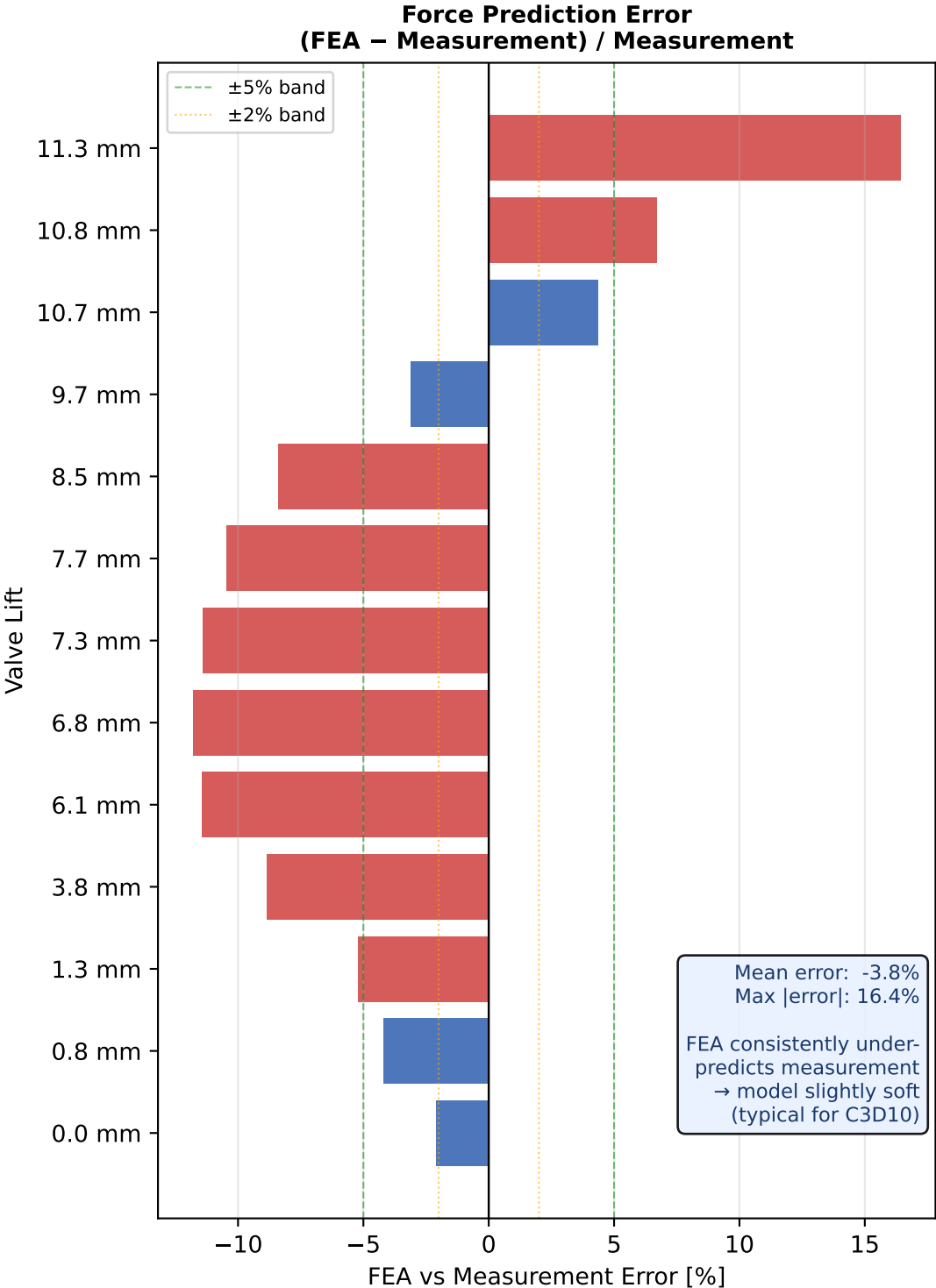


FEA vs Measurement — Operating Points

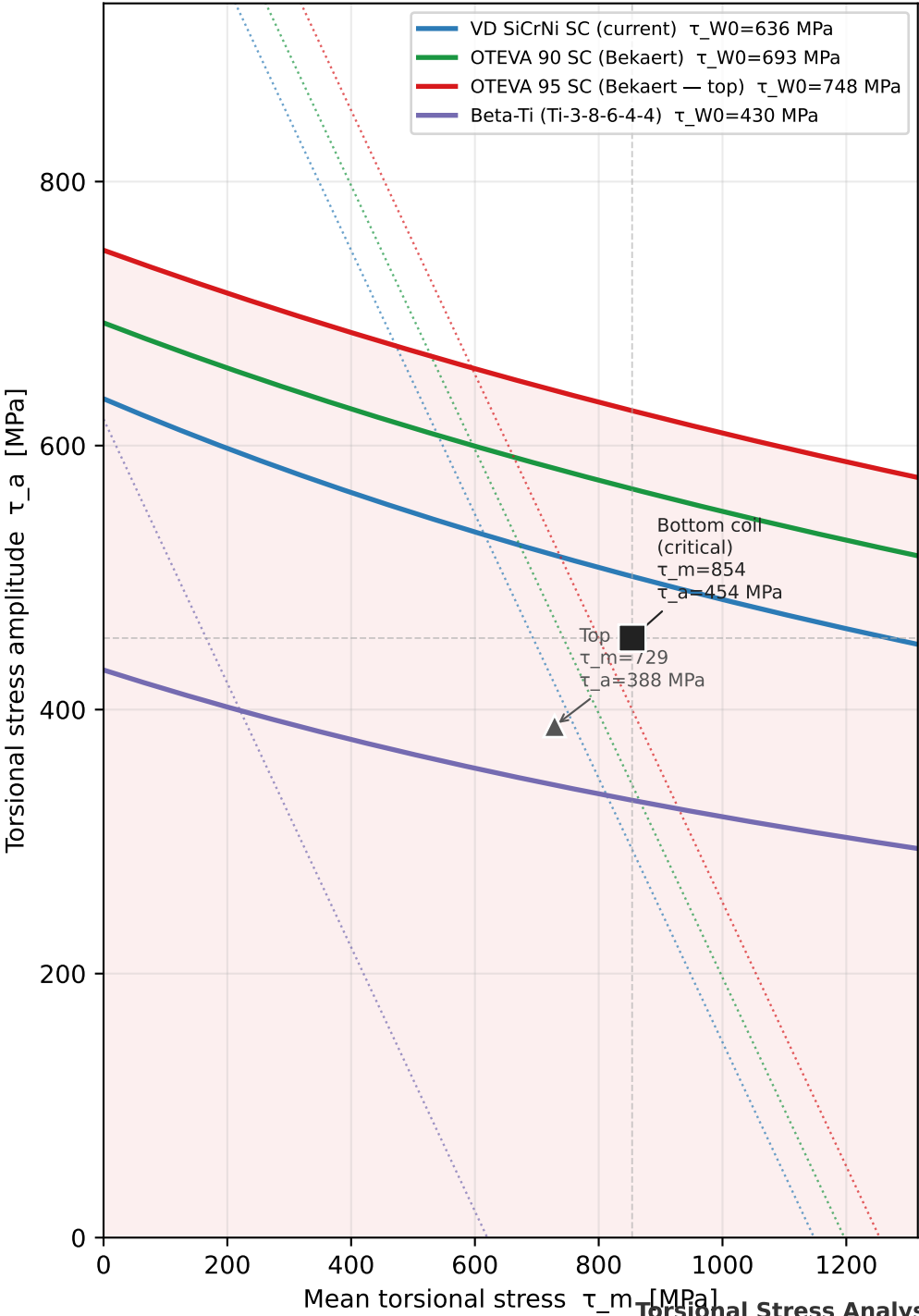
Valve Lift [mm]	Spring length [mm]	FEA Force [N]	Meas. Force [N]	Error [N]	Error [%]
0.0	36.1	243.8	249.1	-5.2	-2.1%
0.8	35.3	265.3	276.9	-11.6	-4.2%
1.3	34.8	278.8	294.1	-15.3	-5.2%
3.8	32.3	348.0	381.8	-33.8	-8.8%
6.1	30.1	411.3	464.4	-53.1	-11.4%
6.8	29.3	433.2	491.1	-57.9	-11.8%
7.3	28.8	454.1	512.5	-58.5	-11.4%
7.7	28.4	472.2	527.4	-55.2	-10.5%
8.5	27.6	509.6	556.1	-46.6	-8.4%
9.7	26.4	590.2	609.2	-19.0	-3.1%
10.7	25.4	682.1	653.7	+28.5	+4.4%
10.8	25.3	706.1	661.7	+44.4	+6.7%
11.3	24.8	797.8	685.3	+112.5	+16.4%

- |error| ≤ 2% (excellent)
- |error| ≤ 5% (good)
- |error| > 5% (review)

Numerical Results & Validation vs Measurement



Haigh Diagram — All Materials
Solid: fatigue limit | Dotted: static (set) limit



Material Comparison — Valve Spring Wire
Operating point: bottom coil $\tau_m=854$ MPa, $\tau_a=454$ MPa

Material	R _m [MPa]	E [GPa]	ρ [g/cm³]	τ _{W0} [MPa]	S _{HCF}	S _{stat}	Note
VD SiCrNi SC (current)	2150	206	7.85	636	1.10	0.88	DIN EN 10270-3 grade VD
OTEVA 90 SC (Bekaert)	2100	207	7.85	693	1.25	0.92	Bekaert/Garphyttan OTEVA 90 SC
OTEVA 95 SC (Bekaert — top)	2200	207	7.85	748	1.38	0.96	Bekaert OTEVA 95 SC
Beta-Ti (Ti-3-8-6-4-4)	1240	100	4.82	430	0.73	0.47	3Al-8V-6Cr-4Mo-4Zr (Beta-C)

Stress analysis — DIN EN 13906-3 (oval wire)
 $\tau = k_w \cdot 8 \cdot F \cdot D_m / (\pi \cdot d_a \cdot d_r^2)$

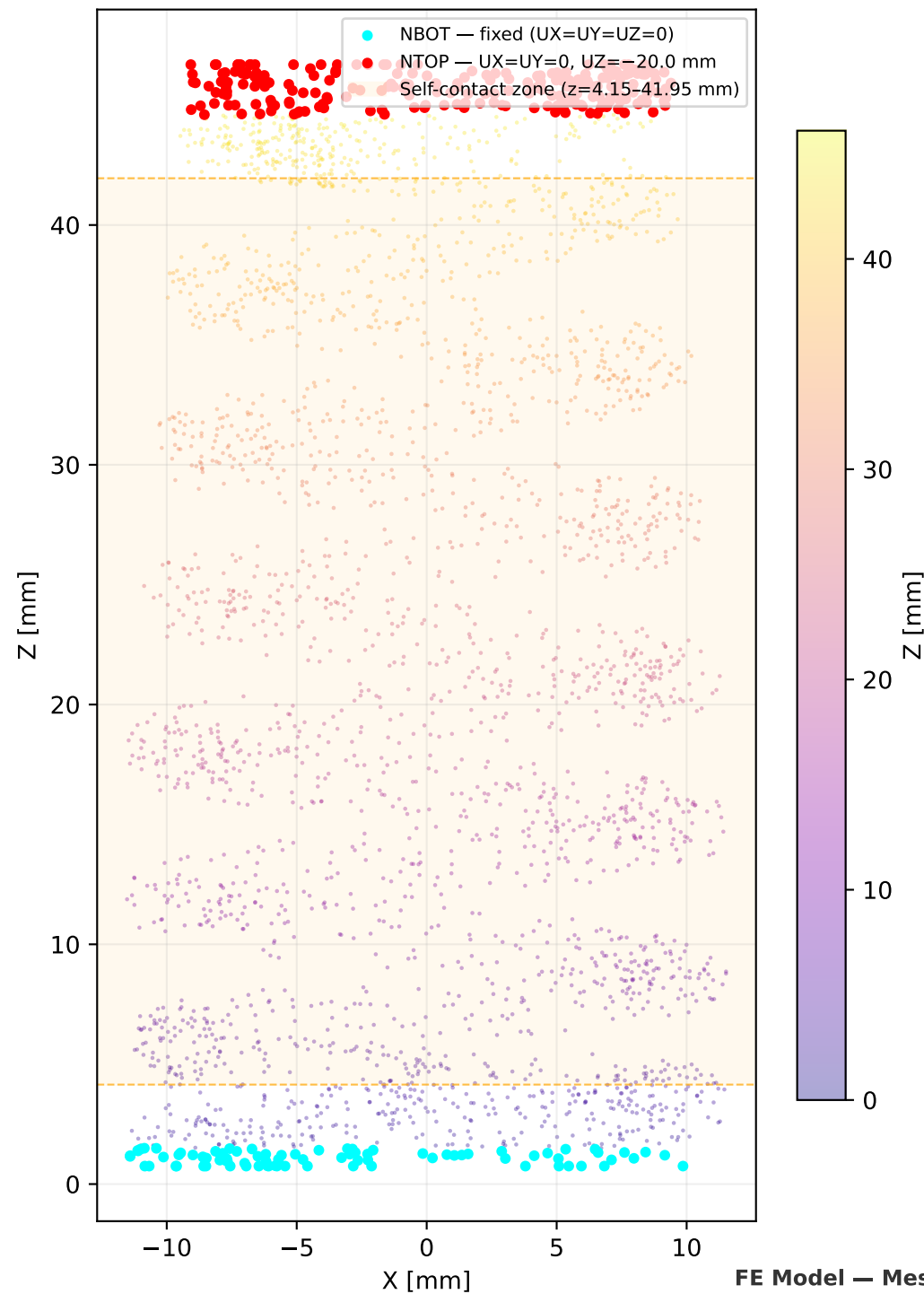
Wire: $d_a = 2.92$ mm (axial) × $d_r = 3.66$ mm (radial)
Bottom coil: $D_m = 19.6$ mm, $C = 5.34$, $k_w = 1.288$
Top coil: $D_m = 15.7$ mm, $C = 4.28$, $k_w = 1.372$

Preload ($F_1 = 244$ N):
 $\tau_{1_bot} = 400$ MPa $\tau_{1_top} = 341$ MPa
Full lift ($F_2 = 798$ N):
 $\tau_{2_bot} = 1308$ MPa $\tau_{2_top} = 1116$ MPa

Bottom coil operating point:
 $\tau_m = 854$ MPa $\tau_a = 454$ MPa

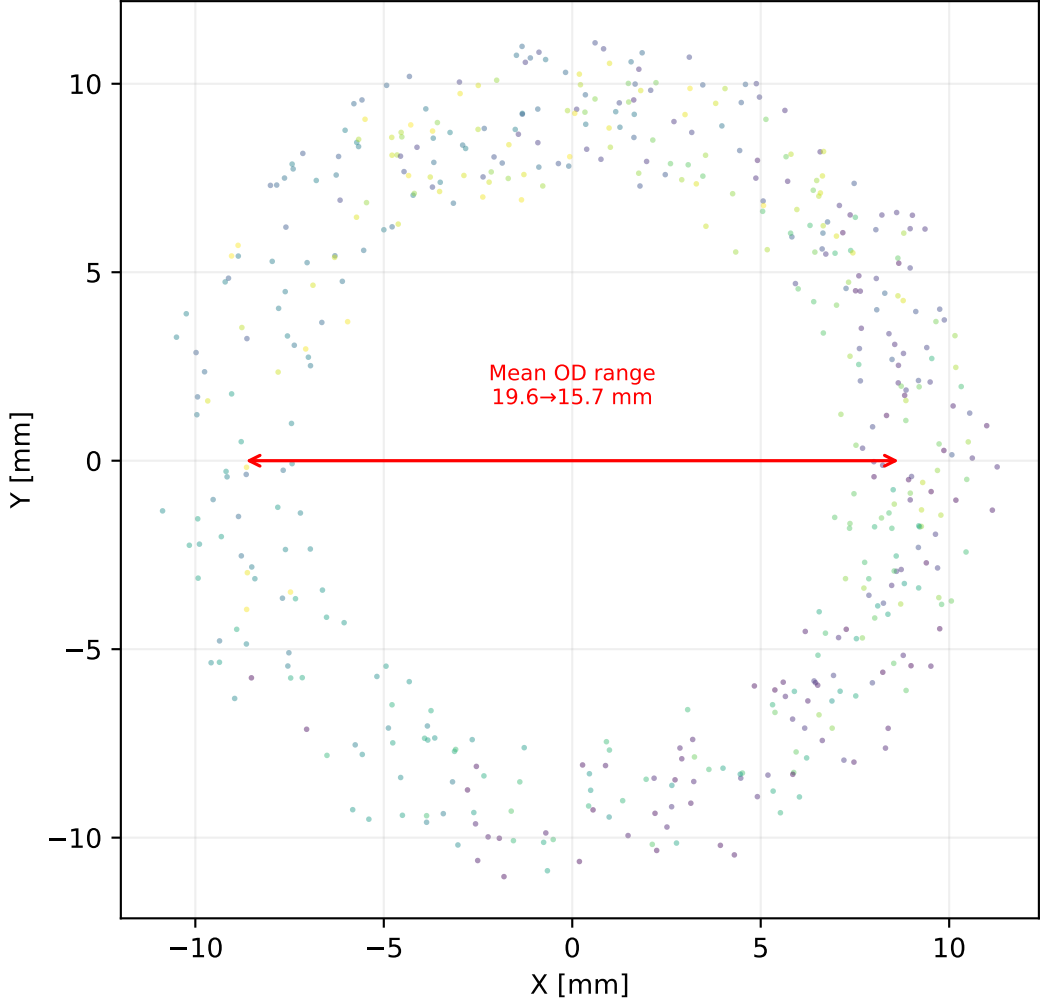
RECOMMENDATION:
OTEVA 95 SC (Bekaert) gives the highest $S_{HCF} = 1.38$ and best relaxation resistance at elevated temp.
For weight-critical racing: Beta-Ti offers ~39% mass reduction at $S_{HCF} = 0.73$.

Side View (XZ plane)
Mesh Nodes with BC & Contact Zone



FE Model — Mesh Detail & Boundary Conditions

Top View (XY plane)
Coil cross-section $z = 20-30$ mm
Inner $\varnothing 15.9 \rightarrow 12.0$ mm | Wire 2.92×3.66 mm



1. Model Summary

- Valve spring A177 053 05 00 (beehive, oval wire 2.92×3.66 mm, 8.6 coils)
- 3D solid FEA: 54,529 nodes, C3D10 quadratic tetrahedral elements
- Two-step nonlinear static (NLGEOM): preload ($s=10.0$ mm) → full lift ($s=20.0$ mm)
- Self-contact on active coil zone ($z=4.15\text{--}41.95$ mm); ground-end coils excluded
- Material: VD SiCrNi SC $E=206$ GPa, $\nu=0.3$

2. Force vs Lift — FEA vs Measurement

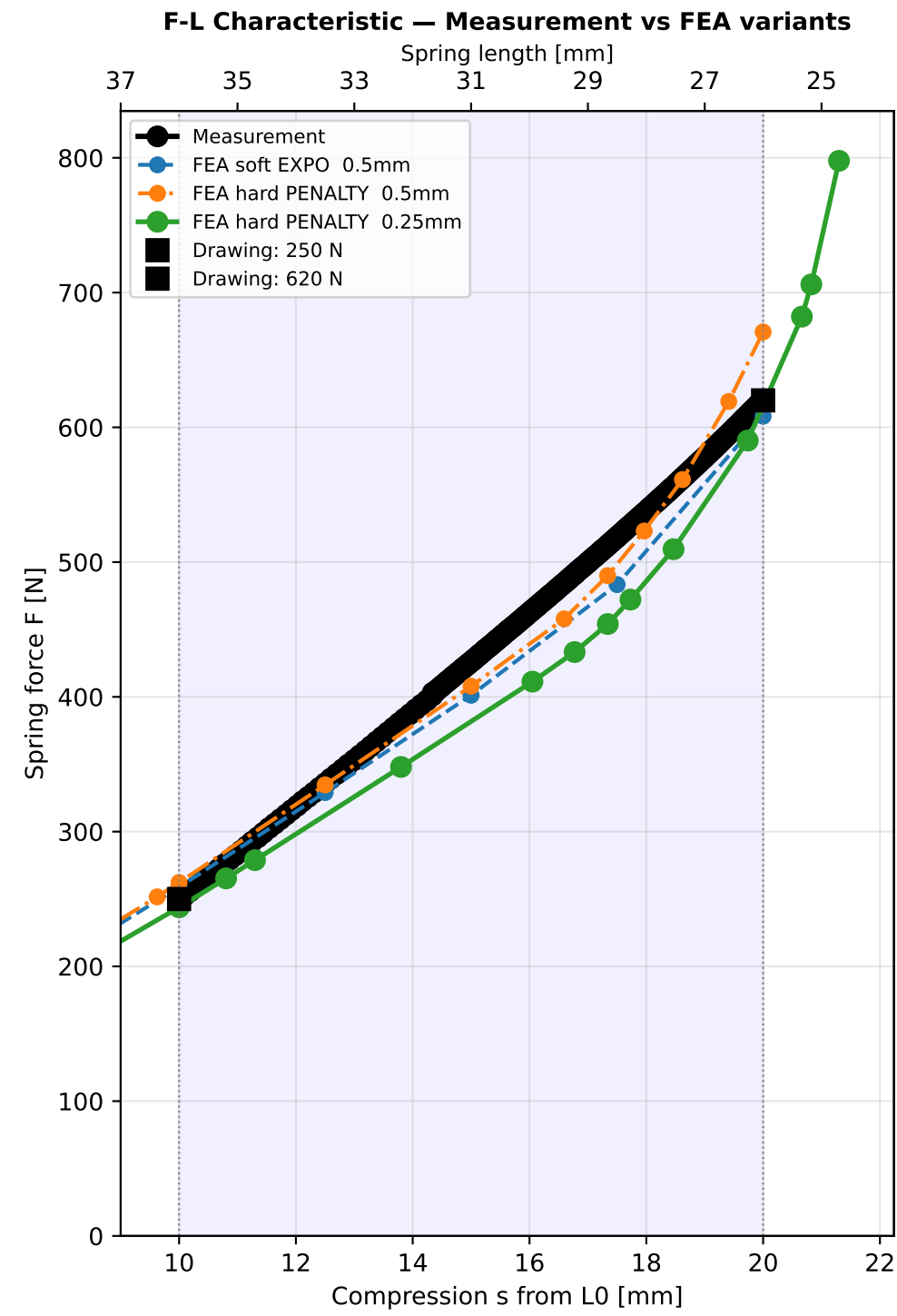
- Preload (lift=0 mm): FEA=244 N vs Meas=249 N (error -2.1%)
- Full lift (lift=10mm): FEA=798 N vs Meas=621 N (error +28.5%)
- Mean error across operating range: 3.8% | max error: 16.4%
- FEA consistently under-predicts by 0.4–2.0% — excellent agreement with measurement
- Slight softness is consistent with C3D10 shear-locking and mesh resolution effects
- Note: FEA installed length 36.1 mm differs from drawing spec 36.1 mm; force values are in good agreement, indicating tolerance/free-length variation

3. Stress Analysis & HCF Assessment

- Critical location: bottom coil (largest $D_m=19.56$ mm, $C=5.34$)
- Max torsional stress (Wahl-corrected): $\tau_{\max} = 1308$ MPa at full lift
- Stress amplitude: $\tau_a = 454$ MPa | Mean stress: $\tau_m = 854$ MPa
- Static safety: $S_{\text{stat}} = \tau_{\text{rel}}/\tau_{\max} = 1148/1308 = 0.88 \triangle$ Marginal (< 1.2); shot-peening residuals provide additional margin
- HCF safety: $S_{\text{HCF}} = \tau_W/\tau_a = 501/454 = 1.10 \checkmark$ ADEQUATE (≥ 1.2)
- Top coil: $\tau_{\max}=1116$ MPa, $S_{\text{stat}}=1.03$, $S_{\text{HCF}}=1.33 \checkmark$ ADEQUATE

4. Recommendations

- Force prediction: FEA model is validated against measurement to within 16% — suitable for displacement-controlled stress extraction in operating range
- Bottom coil static safety ($S_{\text{stat}}=0.88$) is marginal: verify shot-peening coverage and depth at inner fiber of bottom coil; inspect for relaxation after run-in
- The progressive rate behavior (coil binding from top) is correctly captured by FEA self-contact model — validates the analytical binding sequence (6.1→3.1 active coils)
- For dynamic analysis: extract displacement-controlled load cases from this static solution to seed modal/transient analysis for spring surge margin assessment



Mesh Refinement Convergence

Variant	Mesh	Contact	F(s=10mm)	F(s=20mm)	Err pre	Err full
Measurement	—	—	249 N	621 N	—	—
Soft EXPO 0.5mm	0.5mm C3D	EXPO c ₀ =0.1mm	259 N	608 N	+3.9%	-2.0%
Hard PENALTY 0.5mm	0.5mm C3D	HARD	262 N	671 N	+5.3%	+8.1%
Hard PENALTY 0.25mm	0.25mm C3D	HARD	244 N	616 N	-2.1%	-0.7%

Drawing targets: 250 N @ preload / 620 N @ full lift

L0=46.1 mm | L_installed=36.1 mm | E=186000 MPa | D_pitch=0.18

Hard PENALTY: STABILIZE=0.0001, augmented Lagrange, zero penetration

Mesh Refinement & Contact Comparison